

# Using heat

SOME MACHINES HARNESS THE ENERGY IN HEAT TO CREATE MOTION, WHILE OTHERS TRANSFER HEAT TO WARM FOOD.

Many vehicles are powered by harnessing the heat energy released from fuels. By contrast, a fridge releases heat to keep the contents inside cool so they can last longer.

## Internal combustion engine

Most road vehicles are powered by internal combustion engines. In external combustion engines, such as steam locomotives, the burning fuel is kept separate from the high-pressure steam that drives the engine. In internal combustion engines the power comes from burning fuel (gasoline or diesel) inside a cylinder, creating motion in a four-stroke cycle.

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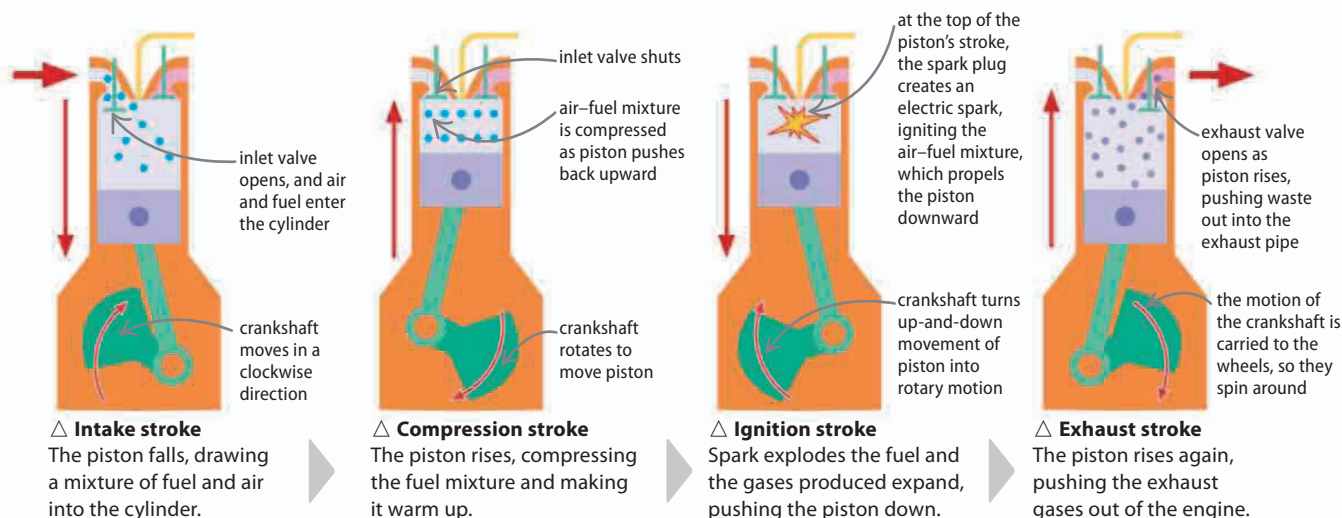
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◀ 188–189 Heat transfer

Power generation

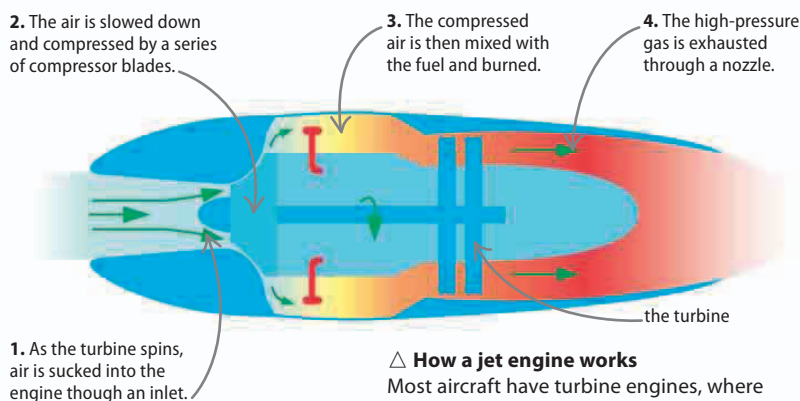
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The first internal combustion engine design was fueled by **gunpowder**.



## Jet engine

Jet engines on aircraft and inside the fastest ships convert heat energy into motion using a turbine. This is a series of propeller-like blades that spin when fast-moving gases flow over them. The gases used are air and the exhaust produced by burning fuel. The spinning turbine drives a compressor, which draws air into the engine and squeezes it so it gets hot. The hot air makes the fuel burn more quickly, driving the turbine around faster. The aircraft is thrust forward by the jet of gas sent backward by the turbine.



### △ How a jet engine works

Most aircraft have turbine engines, where a turbine is used to pull air into the engine, which begins the process described above.